Coordinate methods with quadrilater

1 $(-2, 2)$

2 $(-2, -1)$

3

a $T\,(-7, -2)$

b $p = -9$

c $q = -6$

4

a

i -2 ii $-\frac{1}{2}$ iii -2 iv $-\frac{1}{2}$

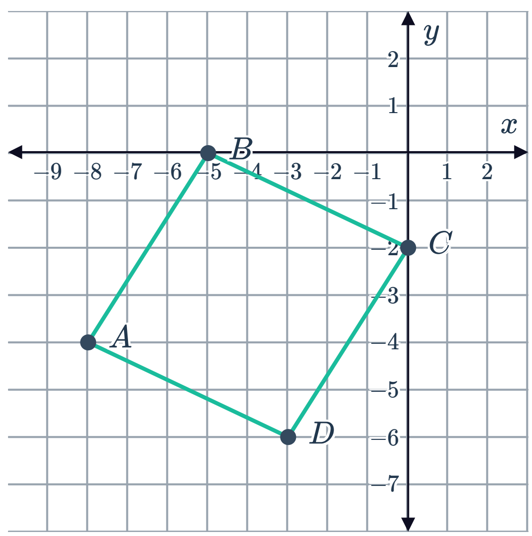
b

i $2\sqrt{5}$ units ii $2\sqrt{5}$ units

c Rhombus

5

a



b

i $\frac{4}{3}$ ii $-\frac{2}{5}$ iii $\frac{4}{3}$ iv $-\frac{2}{5}$

c $\overline{DC}$ and $\overline{AB}$, $\overline{BC}$ and $\overline{AD}$

d A parallelogram

6

a

i $-\frac{3}{4}$ ii $-\frac{3}{4}$ iii $\frac{4}{3}$ iv $\frac{4}{3}$

b

i 5 units ii 5 units

c Square

7

a

i $-\frac{3}{4}$ ii $-\frac{3}{4}$ iii $\frac{4}{3}$ iv $\frac{4}{3}$

b

i 10 units ii 5 units

c Rectangle

Additional questions

8

a No sides are congruent, so all sides have different lengths

b At least two sides are congruent, so at least two have the same length

c All sides are congruent, so have the same lengths

d Two sides are perpendicular, so their slopes will be opposite reciprocals

9

a

i 8 units ii 8 units

iii 8 units iv 8 units

b Yes

c Yes

d Square

10

a

i -1 ii 1

b

- $AC = 5\sqrt{2}$ units
- $BC = 5\sqrt{2}$ units
- $AB = 10$ units

c Right isosceles triangle

11

a

i $m_{\overline{AB}} = \frac{1}{5}$ ii $m_{\overline{CD}} = \frac{1}{5}$ iii $m_{\overline{AD}} = -\frac{5}{3}$ iv $m_{\overline{BC}} = -\frac{5}{3}$

b This quadrilateral can be most precisely classified as a parallelogram.

12

a

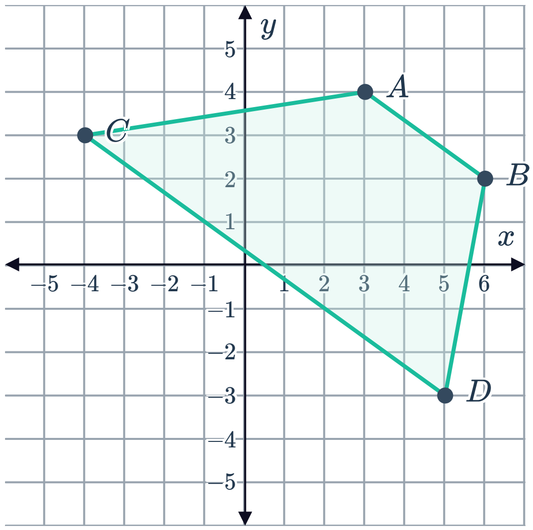
i $BC = 8$ units ii $AB = 2\sqrt{5}$ units

iii $AC = 2\sqrt{5}$ units

13 This quadrilateral can be most precisely classified as a rhombus.

14

a



b Trapezium

15 Parallelogram

16 A right-scalene triangle

17

a

i $m_P = -6$

ii $m_Q = \frac{1}{6}$

iii $m_R = -6$

iv $m_S = \frac{1}{6}$

b Rectangle

18 Given: the vertices

- $A\,(1, -2)$
- $B\,(9, 3)$
- $C\,(7, -1)$
- $D\,(-1, -6)$

Prove: $ABCD$ is a parallelogram

1. Find the midpoints of AC and BD

$$\begin{aligned} M_{AC} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \\ &= \left(\frac{1+7}{2}, \frac{-2+-1}{2}\right) \\ &= \left(4, -\frac{3}{2}\right) \end{aligned}$$

$$\begin{aligned} M_{BD} &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right) \\ &= \left(\frac{9+-1}{2}, \frac{3+-6}{2}\right) \\ &= \left(4, -\frac{3}{2}\right) \end{aligned}$$

2. Show that AC and BD bisect one another
   The definition of a segment bisector is "A segment, ray, line, or plane that intersects a segment at its midpoint." Since $\overline{AC}$ and $\overline{BD}$ have the same midpoints, we can conclude that they intersect at their midpoints. This means they are segment bisectors of one another, or that AC and BD bisect one another.
3. $ABCD$ is a parallelogram
   If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.
   The diagonals bisect eachother, so this is a parallelogram.

19 Quadrilateral $RSTU$ is a rectangle.
Conditions of rectangles state that, "if one angle of a parallelogram is a right angle, then the parallelogram is a rectangle".
Since $\overline{RS} \parallel \overline{TU}$ and $\overline{RU} \parallel \overline{ST}$, $RSTU$ is a parallelogram.
Since $\overline{RS} \perp \overline{ST}$, the parallelogram has a right angle and so it is a rectangle.

20 For $ABCD$ to be a trapezoid, but not an iso[redacted]s trapezoid, we need to show that:

- $\overline{AB} \parallel \overline{CD}$
- $AB \neq CD$

Since we are given the diagram, instead of the coordinates of the points, we can count units on the diagram to help calculate slopes and distances on the coordinate plane.

$$\begin{aligned} m_{\overline{AB}} &= \frac{\text{rise}}{\text{run}} \\ &= \frac{-6}{3} \\ &= -2 \end{aligned}$$

$$\begin{aligned} m_{\overline{CD}} &= \frac{\text{rise}}{\text{run}} \\ &= \frac{-10}{5} \\ &= -2 \end{aligned}$$

$$m_{\overline{CD}} = m_{\overline{AB}}$$

Since we have that $m_{\overline{AB}} = m_{\overline{CD}}$ we can say that $\overline{AB} \parallel \overline{CD}$ and conclude that $ABCD$ is a trapezoid.

$$\begin{aligned} AD &= \sqrt{a^2 + b^2} \\ &= \sqrt{1^2 + 4^2} \\ &= \sqrt{1 + 16} \\ &= \sqrt{17} \end{aligned}$$

$$\begin{aligned} BC &= x_2 - x_1 \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

Since we have that $AD \neq BC$ we can conclude that $ABCD$ is not an isosceles trapezoid.

21 The diagonals of parallelograms bisect each other, so we can first find the midpoint of PQ and then find the endpoint S, such that point T is the midpoint of RS

1. Find T, the midpoint of PQ.
   $T = (-8, -8)$
2. Find the point S, such that point T is the midpoint of RS.
   $S = (-10, -15)$

22 There are many different ways to prove a quadrilateral is a square. Here are two possible methods.
Method 1:

- Find the lengths of all sides to show they are all same, so it has four congruent sides and is a rhombus
- Calculate the slope of the four sides and show that the slopes of consecutive sides are opposite reciprocals, so sides are perpendicular and it has four right angles, so is a rectangle
- Since it is a rectangle and a rhombus it is a square

Method 2:

- Find the lengths of lengths of the two diagonals and show they are same, so the diagonals are congruent and it is a rectangle
- Calculate the slope of the two diagonals and show they are opposite reciprocals, so diagonals are perpendicular and it is a rhombus
- Since it is a rectangle and a rhombus it is a square

Method 2 is more efficient as it requires two distance and two slope calculations, while Method 1 requires four distance and four slope calculations.